External Financing from Dividends:
Evidence from Japanese Business Groups

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Abstract

With a large sample from Japan during the period of 1990-2012, we find firms that belong to business groups ('*keiretsu*') pay more cash dividends than firms not affiliated with any group. The difference between the two groups of firms is greater when the dividend-receiving firms have better investment opportunities, are in financial distress, or when the linkage between them and the dividend-paying firms is stronger. Using exogenous changes to taxes on corporate dividends and difference-in-difference and falsification tests, we further establish that keiretsu firms' dividend payouts have a causal impact on receiving firms' investment and debt policies. Our results highlight the importance of inter-firm financing, especially during periods of high external financing costs, even for firms in a developed economy.

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KEY WORDS: Japan, business group, cross-holding, dividend, investment, debt.

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1. Introduction

Fazzari, Hubbard and Petersen, 1988..

One of the important lessons from the 2007-2009 global financial crisis is the fragility of the financial system and its impact on corporations. During periods of boom, financial markets and institutions have ample funds and liquidity to distribute among corporate and household sectors, and, corporations can finance their investment projects at low costs. During periods of crisis, however, there is severe contraction in available credit and funds in the system and such credit and capital crunches have significantly adverse impact on firms' operations including capital budgeting policies. How do firms cope with the frequent rise and fall in the markets and financial institutions and design a 'smooth' plan for investments and growth?

A growing strand of literature studies business groups—a dominant organizational form outside the US (e.g., La Porta, Lopez-de Silanes, Shleifer, 1999), and find that firms within the groups can partially *internalize* the cycles and costs in external financing. They achieve this through business networks, especially during periods of crises and for firms that face high costs of external financing. There are different forms of financing to and from the networks, including trade credits which have been studied extensively. Different from most prior studies, in this paper we look at Japan, one of the most developed countries with advanced markets and financial institutions. We distinguish two groups of firms—firms that belong to business groups, or *keiretsu*, vs. those that do not.

Our motivation stems in part from Dewenter and Warther (1998), who indicate that cash dividends are a possible mechanism for the inter-corporate transfer of resources within keiretsus. We examine two related questions. First, we are interested in whether keiretsu firms' payout policies are related to *other* keiretsu firms' financing needs. Second, we study whether dividend-receiving firms' ('receivers') investment policies are related to dividends from the dividend-paying firms ('payers') within the same group; we also study whether receivers' leverage ratio is affected by dividends received from the payers. Moreover, we use changes to taxes on corporate dividends as the main identification strategy to try to establish that keiretsu firms' dividend payouts have a *causal* impact on receivers' investment and debt policies. In all of these tests, non-keiretsu firms serve as the natural control group.

Our panel data set includes 2,521 firms for the period 1990-2012. We identify 555 firms belonging to one of the six business groups following the methodology of Brown and Fung (2009) and the *Industrial Groupings in Japan—the Anatomy of the Keiretsu* (Brown & Company 1999, 2001). In order to calculate cash dividends received by keiretsu and non–keiretsu firms from a keiretsu firm, we identify the name and percentage shareholding of the top ten shareholders for the keiretsu firms.[1] Investment is estimated as the annual change in total assets of a firm, as well as annual changes in research and development (R&D) expenditures and changes in capital expenditure (CAPEX).

In our first set of tests, we find that the average payout for a keiretsu firm is US$ 6.36 million per year (1990 prices), 45% more than non-group affiliated firms; large, nucleus keiretsu firms with strong financial conditions pay out even greater amounts. After incorporating a set of firm controls and determinants of dividend policies, we find that the keiretsu payout premium over non-keiretsu firms remains large at 22%. The payouts are also greater relative to non-group firms, when the receivers within the group have better investment opportunities (as proxied by higher market-to-book ratios and industry-level investments), are in financial distress, or when they have a closer tie to the dividend payers as proxied by the size of shareholdings. By contrast, we do not find similar patterns in payouts among non-keiretsu firms.

Next, we examine whether and how firms' investment and leverage depend on cash dividends received from other firms. In the panel regression models we include firm and year fixed effects to absorb the effects of macroeconomic conditions and all time-invariant factors on investment and leverage. We find that for keiretsu firms, a greater amount of cash dividends received is associated with higher investment and lower leverage over the next two subsequent years. This relation strengthens as the receivers' tie to the payer strengthens.

It is possible that we may not have fully accounted for all the factors that may lead to correlations between dividends received and keiretsu firms' investment and leverage decisions. In order to establish causal impact of dividends received on future investment and leverage, we exploit an exogenous shock in taxes on corporate dividends—taxes on cash dividends (paid by dividend recipients) dropped from 43.6% to 10% in 2003, a substantial fall—and see whether

[1] Following prior work (e.g., Wu and Xu, 2005, and Brown and Fung, 2009), we make the assumption that the percentage shareholdings among firms are stable over time.

dividend payouts are associated with different investment and debt policies for keiretsu and non-keiretsu firms; i.e., we conduct difference-in-differences tests around the tax cut year. We find that while both types of firms increased payouts following the tax cut on dividends, the increase in keiretsu firms' payouts is 14.1% greater than that for non-keiretsu firms during the post-tax cut period of 2004-2012. Further, if firms belong to the same keiretsu are in financial distress, dividend paying firms within the same business group raise their payouts by an additional 11.1% after the tax cut.

Turning to firm investment, as measured by annual changes of book assets (Hoshi, Kashyap and Scharfstein, 1990), keiretsu firms exhibit an additional 0.87% annual increase in investments over the next two years relative to non-keiretsu firms, per 1% increase in dividend received, during the period 2004-12. The estimate increases by a further 0.79% if dividend-receiving firms in the same business group are in financial distress. We therefore show that the impact of dividend received on future investments in keiretsu affiliated firms is substantively larger than its influence in nonaffiliated firms. We find similar results on the differential impact on firms' leverage ratios. When dividends received increase by 1% in the current year, keiretsu firms' leverage ratio drops by 0.92% more annually, over the next two years, as compared to non-keiretsu firms, during the period 2004-12. The estimate is larger by an additional 0.55% if dividend-receiving firms in the same business group are in financial distress. These results show that dividend receiving firms use the cash proceeds from payers within the group to pay down debt more aggressively than their counterparts unaffiliated with any business group.

To strengthen the argument that dividend payouts have a causal impact on keiretsu firms' investment and debt policies, we also conduct a difference-in-differences test around a period of no known shocks on tax rates, *i.e.*, a falsification test. We focus on the period 2004 to 2012 – a period of a fixed dividend tax rate of 10%, and specify a fictional tax rate alteration year of 2006. We reassuringly find that there is no significant change in the amount of payout of keiretsu firms relative to non-keiretsu firms during this period and no significant change in the relations between dividend received and investment or between dividends and leverage ratio.

Our results extend the literature on the role of business groups by highlighting the importance of inter-firm financing in the form of dividends for firms in a developed economy. Gopalan, Nanda and Seru (2014) employ a multi-country dataset and find that firms within the same business group distribute dividends when affiliated firms have investment opportunities that require (external) financing. Our results on the difference between keiretsu firms vs. non-keiretsu firms from Japan confirm their results. More importantly, our identification based on an exogenous shock to the tax rates on dividends, along with the falsification tests, allows us to demonstrate causal impact of dividend from keiretsu firms on affiliated firms' investment and debt policies.

Our paper also extends the strand of literature studying Japanese business groups. Prior research has shown that through cross-shareholding, the ties connecting group firms are both financial and non-financial depending upon the level and type of affiliation with the group. For example, Pinkowitz and Williamson (2001) show that the financial ties among group firms are primarily in the form of loans from the bank (or other large financial institutions), typically sitting at the top of the hierarchy, supplemented by intra-group debt. These connections are particularly valuable for group firms in financial distress (e.g., Hoshi, Kashyap and Scharfstein, 1990). We identify a specific financial channel—dividend, through group firms help each other, and show that the proceeds from dividends received provide important source of financing for firms to fund investment and to pay down debt.

The rest of the paper is organized as follows. Section 2 reviews institutional background information and develops our hypotheses and empirical methodologies. Section 3 presents the data and summary statistics, and Section 4 presents empirical results. Finally, Section 5 concludes. The Appendix contains explanations of all the variables used in the paper.

2. Institutional Background Information, Hypotheses and Methodology

Our point of departure is the well-established notion that firms affiliated with a keiretsu group are expected to receive help from the main bank and other firms in the group in times of need and reciprocate such assistance. As pointed out by Berglof and Perotti (1994), the cross-holdings of debt and equity in keiretsu affiliated firms is a governance mechanism that supports cooperation and mutual monitoring. Firms assist each other to stabilize relations with customers and suppliers and facilitate long-term planning. For instance, Japanese keiretsu firms have protected each other from (external) takeover threats via allegiances and cross holdings of stocks and bonds.[2]

2.1 Testable Hypotheses

What we examine in this paper, which has not been studied previously, is whether keiretsu firms' dividend payouts can have a causal impact on keiretsu dividend receiving firms' investment and leverage policies.

The alternative financing mechanism of an intra-group investment channel (e.g. a loan contract) directly links the cash rich firm to the firm with the investment opportunity or impaired financial status. Such an intra-group investment channel may prove attractive due to costs (such as levied taxes) associated with the distribution and reinvestment of dividends. It may also, however, prove unattractive due to the imposition of an obligation to repay, which may in part defeat the purpose of assisting an affiliated firm. Moreover, Gopalan, Nanda and Seru (2007) report that potentially costly auditing covenants, with respect to inter-corporate investments, are routinely found in loan contracts in India. This is not surprising given the opacity of this investment channel. Non-keiretsu affiliated firm shareholders may, in addition, observe the intra-group investment and discipline management for disbursing cash preferentially to keiretsu member firms, by increasing the cost of equity. Dividends, in contrast intra-group investment channel, are distributed across all shareholders, and may hence enable affiliated firms to reallocate capital across a business group. Hence, it is an empirical question of the relative roles of these finance channels. Indeed, when we estimate the influence of dividends received on recipient firm capital budgets and financial leverage, we account for the influence of alternative sources of capital including the issuance of equity, long-term debt and trade credit and receivables.

We test two main hypotheses. Our first main hypothesis test is in relation to a higher dividend payout in keiretsu firms, which is due, in significant part, to same keiretsu group firms' investment opportunities, especially in settings of firm financial distress. Our second main hypothesis is in relation to whether the dividend received by keiretsu firms is, indeed,

[2] Brown and Fung (2009), however, report reduced allegiances in the market for corporate control in Japan, which they associate with a raft of legislation which culminated in the New Corporation Law in 2006.

an important determinant of their investment or financial leverage policies, with a particular focus in settings of recipient firms in financial distress. If managers of Japanese keiretsu affiliated firms use cash dividends to finance their investment or the financial leverage policies of other keiretsu firms, then this constitutes a new influence on the cash dividends of business group members in a developed economy.

Tests of keiretsu firms' dividend pay-out determination

Accounting for the effects of well-known determinants of dividend pay-out, experienced by firms in a competitive market, our first major hypothesis suggests that the management of keiretsu affiliated firms is expected to pay out more than the pay out of non keiretsu affiliated firms.[3] Accordingly, keiretsu affiliated firms are subject to the same determinants of dividends as unaffiliated firms as well as an additional influence of business group membership. Thus, our first major hypothesis can be stated:

H_1: *There is a larger amount of pay-out, accounting for well-known factors associated with dividend determination, from keiretsu affiliated firms than from non-affiliated firms.*

To test this hypothesis, we specify panel regression models and incorporate a dummy variable, *KRF*, which takes a value of one if a firm is keiretsu affiliated and zero otherwise. We estimate model specifications that are variants of the following form:

$$Dividend_{i,t} = \alpha_0 + \gamma_0 KRF_i + \sum_{j=1}^{n} \beta_j \, CV_{s,t-1} + \alpha_{ind} + \alpha_t + \varepsilon_{it}, \quad [1]$$

where *Dividend* refers to the natural logarithm of common cash dividends, i refers to the i^{th} individual firm, *t* refers to the year (year effects) and *ind* (industry effects) refers to the industry of observation. CV_s refers to a specific control variable *s*. Of particular interest, is the dividend pay-out premium, γ_0, which estimates the additional dividend pay-out of keiretsu affiliated firms, relative to non-affiliated firms. We expect the dividend pay-out premium, γ_0, to be positive $\gamma_0 > 0$.

To further investigate this main hypothesis, we test if the dividend pay-out premium, of the

[3] We use a flexible specification which includes a broad set of dividend determinants consistent with Allen and Michaely (2003), DeAngelo, DeAngelo and Stulz (2006) and Chay and Suh (2009).

dividend paying firm, is independent to the strength of its keiretsu affiliation – weak (within group ownership is less than 50%) or strong (within group ownership is more than 50%), and if the pay out premium is independent to whether the dividend paying firm is a main keiretsu bank. We expect that with an increasing strength of the keiretsu affiliation there should be an increase in the size of the keiretsu firm dividend pay-out premium. The stronger the keiretsu affiliation, the greater the amount of a dividend pay-out which will be received by keiretsu firms, and, we expect, the greater the concern of the dividend paying firm for the capital budget, financial leverage and financial status of keiretsu affiliated firms.

In order to better understand the determination of the keiretsu dividend pay-out premium, we conjecture that the investment opportunities of the keiretsu dividend receiving firms will have an increasing influence on dividend pay-out, the more closely affiliated the keiretsu status of the dividend receiving firm. We allow the dividend receiver status to be a member of the same keiretsu group, a member of a different keiretsu group or a non-affiliated firm. We expect that the more closely related the status of the dividend recipient firm, relative to the dividend paying firm, and the greater its investment opportunities, the larger the dividend pay-out premium.

A closely related sub-hypothesis, with respect to the investment opportunities of keiretsu dividend receiving firms, as a determinant of the keiretsu dividend payout premium, is a test of the independence of the status of the dividend receiving firm *and* the strength of the keiretsu affiliation of the dividend paying firm, in respect to the determination of the keiretsu firm pay out premium. We expect that the stronger the keiretsu affiliation of the dividend payer, the more closely related the status of the dividend recipient firm and the larger its investment opportunities, the larger the dividend pay-out premium.

One further reason keiretsu affiliated firms can pay-out more, is that other keiretsu firms may experience financial distress, and there is an incentive – within keiretsu groups – to assist affiliated firms experiencing financial distress.[4] The financial distress of the keiretsu dividend

[4] Hoshi, Kashyap and Scharfstein (1990) suggest that keiretsu group main banks preferentially lend to financially distressed keiretsu affiliated *and* non-affiliated firms, which have strong financial ties to the main bank. These strong financial ties are understood to reduce informational asymmetries and agency conflicts. Overall, keiretsu affiliated financially distressed firms are shown to perform better than non-affiliated firms, which are financially distressed. Hence, affiliation with a keiretsu group can facilitate

receiving firms is expected to have an increasing influence on dividend pay-out, the stronger the keiretsu status of the dividend receiving firm. We allow the dividend receiver keiretsu affiliation status to be a member of the same keiretsu group, a member of a different keiretsu group or a non-affiliated firm. A closely related sub-hypothesis, with respect to the financial distress of keiretsu dividend receiving firms as a determinant of the keiretsu dividend payout premium, is a test of the independence of the strength of the keiretsu affiliation of the dividend payer (weak, strong or main bank) and the keiretsu affiliation status of the financially distressed firm. It is expected that the stronger the keiretsu affiliation of the dividend payer and the more closely related the keiretsu affiliation status of the dividend recipient firm experiencing financial distress, the larger the dividend payout premium. Finally, an identical set of hypotheses tests is conducted, where the fraction of ownership held by a financially distressed keiretsu firm, of a keiretsu affiliated dividend paying firm is used, instead of a dichotomous variable, to indicate the dividend receiver keiretsu affiliation status.

Tests of role of dividend received in keiretsu firms' investment and finance decisions

If keiretsu firm pay out policies are linked to the investment or financial leverage policies of other keiretsu firms, we can expect keiretsu firm dividend received to be positively associated to same keiretsu firm investment (changes in the book value of total assets, research and development expenditure and capital expenditutre) and negatively associated to same keiretsu firm financial leverage policies, to a greater extent than the association exhibited by unaffiliated firms.[5] We therefore test whether keiretsu firms exhibit a significantly larger association between cash dividends *received* and leading firm level investments and financial leverage, than unaffiliated firms.[6] Hence, our second major hypothesis test is:

H_2: *The impact of dividends received, on firm level leading investment variables and finance, will be larger in keiretsu firms than in non keiretsu firms.*

[5] Fazzari, Hubbard and Petersen, (1988) have shown that any financially constrained dividend recipient may use new cash holdings from dividends to realise its capital budget for investment.

[6] Our cash dividends received data are estimated via cross shareholdings. Specifically, the percentage holdings of the top ten shareholders of keiretsu firms is sourced in the Industrial Groupings in Japan (2001) book. The Industrial Groupings in Japan publishes data observed in 2000, corresponding to the mid-point of our full sample period. As a result of a focus on the top ten shareholders exclusively, however, we have a lower bound estimate on net dividends received. This is, therefore, a conservative estimate of the amount of dividend received and the number of keiretsu (and non-keiretsu) net dividend receivers. Further detail is provided in appendix A1.

Our model specifications are variants of the following form:

$$\sum_{l=1}^{2} D_{i,t+l} = \alpha_i + \partial_0\, DR_{i,t} + \partial_1\, DR_{i,t} * KRF_i + \sum_{j=1}^{n} \beta_j\, CV_{i,t} + \alpha_t + \varepsilon_t \qquad [2]$$

We, thus, study the determination of leading (the summation of the subsequent two periods') firm level decisions, *D*, in respect to investment, *I*, and leverage, *L*. Consistent with Hoshi, Kashyap and Scharfstein (1990), Biddle, Hilary and Verdi (2009) and Gopalan, Nanda and Seru (2014), we include a set of suitable control variables, *CV*, in respect to the determination of investment, in our model specification.[7] When we focus on financial leverage rather than investments, our set of control variables is consistent with Flannery and Rangan (2006) **(??)**. Our specifications control for firm fixed effects, α_i, and year effects, α_t. In this regression specification, KRFi, refers to the keiretsu affiliation status of the dividend receiving firm, whether unaffiliated, weakly or strongly affiliated or a main bank.

We include, as a point of focus in the model specification, the natural logarithm of actual cash dividends received, *DR*, by keiretsu affiliated and non-keiretsu affiliated firms, from keiretsu affiliated firms. In comparison with unaffiliated firms, we expect a relatively high association, ∂_1, between dividends received by keiretsu firms and their lead firm level investment and financial leverage variables. This test directly relates to the strength of the keiretsu affiliation (weak or strong affiliations or the main bank of the group) of the firm receiving the dividend. If a dividend receiving firm has stronger keiretsu affiliations, we would expect a stronger association between the keiretsu dividend recipient firm's dividends received and same firm lead investment and financial leverage. Such a finding would be consistent with pay out in keiretsu firms as a pre-determined facilitator of other keiretsu group member firm investment and financial leverage decisions, and, hence, it can in part account for the keiretsu firm dividend pay-out premium.

One implication of having debt is that any earnings generated by new investment projects are appropriated in part by existing debt holders. Thus, excessive debt ('debt overhang') is an impediment to pursuing positive net present value investments (Myers, 1977), due to a concern of an inadequate residual return on investment to equity holders, once debt convenants are honoured. Motivated by the 'debt overhang' hypothesis, we therefore test if firm leverage generally exhibits a negative association with lead investment.

A related hypothesis test, which we conduct, is whether keiretsu affiliated firms invest more or less than their unaffiliated counterparts, and, in particular, whether this difference is altered in periods of financial distress. Hoshi, Kashyap and Sharfstein (1990), it is expected that keiretsu affiliated firms will invest more than unaffiliated firms, and that this association would be strengthened during periods of financial distress.

Taking the 'debt-overhang' hypothesis implication, for the relation between leverage and investment, together with the capacity of affiliated firms to invest more during times of financial distress, we conduct a new hypothesis test. We test if in our sample of keiretsu affiliated firms, firms will rely relatively heavily on cash dividends received to finance their capital budgets when in a state of financial distress. Fazzari, Hubbard and Petersen (1988) have shown that any financially constrained dividend recipient may use new cash holdings from dividends to realise its capital budget for investment. Such financially constrained firms would have limited access to competitive debt markets and a high cost of debt, thus rendering an intra-group capital market more attractive.

PROBABLY SUPPRESS: We also design additional sub-hypotheses test, following Almeida, Hsu and Li (2013), to assess if as information asymmetries become larger (extra-group capital raising opportunities become less attractive) does the financing mechanism through the business group dividends channel become more important with respect to determining future investment. We also test if this latter information asymmetry influence on keiretsu firm financing investment by way of dividends becomes more evident the stronger the keiretsu affiliation of the dividend receiving firm.

Difference-in-differences and Falsification tests in respect to major hypotheses

Evidence in support of the conjectured relations between keiretsu group firms' pay out and keiretsu dividend receiving firms' investment and leverage policies, arising from the tests of our stated main hypotheses and our sub-hypotheses, would be intriguing and could motivate further work. These stated hypotheses tests, however, while suggestive of associations, cannot establish a case for a causal link between keiretsu firm capital budgets and leverage and the dividend pay outs of keiretsu group firms. As a result, we conduct difference-in-differences tests, with a view to better testing the causality inferred with respect to the conjectured relations. To conduct these tests, we avail of two types of exogenous shocks: changes in the cost of external financing and changes in the taxation rate on cash dividends in Japan. In 1996, there was an end to a period of high interest rates on debt in Japan, which commenced prior to 1990. In 1999 there was a tax rate change in respect to cash dividends, from 35% to 43.6%. In 2003, there was a relatively large reduction in the cash dividend tax rate from 43.6% to 10% until end of 2012.[8]

Keiretsu firms' pay-out determination

In order to further test for the presence of a causal link between keiretsu investments, of keiretsu firms which receive cash dividends, and the keiretsu pay out of group member firms, we examine the influence of the cost of external finance and cash dividend tax rate changes as exogenous shocks to the firm cash dividend pay-out decision.[9] If the tax rate rises (falls) as it did in 1999 (2003) we would expect a larger decline (increase) in cash dividend pay-out in keiretsu affiliated firms, relative to the decline (increase) in cash dividend pay-out in non-affiliated firms. To conduct these hypotheses tests, we use variants of the Equation 1 regression specification which can be stated:

$$Dividend_{i,t} = \gamma_1 \mathrm{ES} * KRF + \sum_{j=1}^{n} \beta_j \, CV_{s,t-1} + \alpha_i + \alpha_{ind} + \alpha_t + \varepsilon_{it}, \qquad [3]$$

[8] It is also noteworthy that, in 2003, that there is a major revision of the Corporate Reorganization Law in Japan to allow for Chapter 11-type turnarounds. This may weaken the strength of affiliations which hold together keiretsu groups and thus diminish any intra-group capital market incentive to pay dividends.

[9] The provision of finance by dividends in Japan can be quite costly as each time a dividend is paid and reinvested there is an associated tax levied. However, the shareholder relief tax regime in Japan potentially permits reduced tax rates on dividends received and the exclusion of a proportion of dividend income from taxation (p.6, Becker, Jacob and Jacob, 2013). For instance, tax losses carried forward (see e.g. Becker, Jacob and Jacob (2013), Jacob and Jacob (2013)) can shelter the income of large firms (requires specific book keeping records and documentation to the Director of the District Tax Office) from tax for up to 7 years after the loss is incurred. There is also a standard dividend tax credit rate available to corporate dividend recipients in Japan. It is expected, however, that while this shareholder relief taxation regime reduces the economic burden of double taxation, taxation can still act as a material impediment (at least until the reduction in the taxation rate on dividends received in 2004) to the distribution of cash within a business group by way of the dividend channel. One exception is that the economic burden of double taxation is expected to be slight for firms which experience financial distress as they are less likely to be subject to a material effective taxation rate on cash dividends received.

the new interaction variable, $ES * KRF$, includes, *ES*, which is a dummy variable that takes a value of 1 during the period of the exogenous shock, and is zero otherwise.[10] In our difference-in-differences hypotheses tests, we investigate if the pay out change during periods of relatively expensive external financing, *i.e.* before 1996, and in periods of low dividend taxes, *i.e.* post 2003, is larger in keiretsu affiliated than unaffiliated firms. It is expected that such periods of relatively expensive external finance/ low dividend tax rates would increase the incentive of the management of keiretsu affiliated firms to seek finance within the keiretsu group, by way of, for example, cash dividends, should intra-group cash dividend financing be available. Our estimate of the change in dividends of group member firms in periods of relatively high external borrowing costs / low taxation rates on cash dividends is γ_1. We expect $\gamma_1 > 0$. Finally, we also test whether the magnitude of the estimated difference-in-differences is linked to the investment opportunities or state of financial distress of same keiretsu group affiliated firms, which receive dividends from same keiretsu group firms.

Keiretsu firms' capital budget and leverage determination

During periods of relatively expensive external financing (pre 1996) and low taxation rates on dividends (post 2003), we investigate, using difference-in-differences hypotheses tests, if there is a different impact of cash dividends received, across affiliated keiretsu and unaffiliated firms, on lead firm level investment and financial leverage policies. We expect that the impact of dividends received is larger in keiretsu affiliated than in keiretsu unaffiliated firms, and that this difference grows in periods of expensive external financing and low taxation rates on dividends. To conduct these hypotheses tests, we use variants of the Equation 2 specification which can be stated:

$$\sum_{l=1}^{2} D_{i,t+l} = \alpha_j + \partial_1 \mathrm{ES} * DR_{i,t} * KRF + \sum_{j=1}^{n} \beta_j CV_{i,t} + \alpha_t + \alpha_i + \alpha_{ind} + \varepsilon_t, \qquad [4]$$

It includes a new interaction variable, $\mathrm{ES} * \mathrm{DR} * KRF$, where ES is a dummy variable which takes a value of 1 during periods of exogenous shocks, and is zero otherwise. We expect $\partial_1 > 0$.

[10] These difference-in-differences tests are accommodated in Equation 3 by substituting TY_1999–03, TY_2004–12 or HCBY_1990-96 as exogenous shocks, *ES*. Further detail of these exogenous shocks is presented in Appendix A1.

The ‘debt overhang’ hypothesis suggests that firms with high debt levels prefer not to invest as payoffs from investments will largely go to service debt (Myers, 1977). Equityholders would not be expected to get a high enough return on investment. We therefore also test if the dividend received by financially distressed keiretsu affiliated firms has a more positive (negative) influence on lead investment (leverage), during periods of high external borrowing costs or low taxation rates on cash dividends, than in financially distressed unaffiliated firms.

Falsification tests

Finally, we conduct falsification difference-in-differences tests around a date, where there is no exogenous shock identified. We perform these tests to better understand whether we can infer a causal impact between keiretsu cash dividend pay-out policies and the investment and financial leverage policies of other keiretsu affiliated firms. As a consequence of no *de facto* exogenous shock there should be no relative difference, between keiretsu and non keiretsu affiliated firms’ pay out and investment/ leverage policy changes, at the imposed fictional exogenous shock date. We conduct these tests about a fictional exogenous shock in 2006, between the 2003 to 2005 sub-period and the 2006 to 2007 sub-period.

3. Data

3.1 Sample construction

Our data, which extends from 1990 through to 2012, is sourced in the Worldscope and the Datastream databases.[11] The data is tailored such that it excludes American Depositary Receipts and foreign firms. The sample also excludes firms whose dividends are greater than their total sales, firms whose dividend, net income or sales figures are omitted and firms with negative book value of equity, market-to-book ratio, sales and dividends. In addition, we search the databases for active as well as dead and suspended listings in order to avoid survivor bias. Otherwise, the sample comprises every firm headquartered in Japan which is also listed on the Tokyo and Osaka Stock Exchange – which now have been merged into Japan Exchange Group, for which there is available a set of proxy explanatory variables for at-least five consecutive

[11] All nominal values are adjusted to real values in the year 1990 by using the local consumers’ price index. Our data are detailed in appendix A1.

years. Taking these filters together yields 2,465 firms, of which 555 firms are affiliated to the six Japanese keiretsu.

3.2 Keiretsu affiliation

A distinguishing feature of Japanese capital markets is the existence of industrial groups called keiretsu, these groups constitute distinctive forms of corporate control in Japan. We study horizontal keiretsu groupings where member firms are in different industries, have close ties with a main bank and exhibit a reciprocity in trade and financial relations.[12] For this study, we must distinguish those firms that are members of keiretsu from those that are independent. The classification process is not straightforward (e.g., Miwa and Remseyer, 2001). Following Brown and Fung (2009), the keiretsu classification is obtained from Industrial Groupings in Japan — The Anatomy of the Keiretsu (2001). Although our sample extends from 1990 to 2012, we use the 2000/01 edition to classify our data as this is the most recently available data with respect to the keiretsu group affiliation. This approach is based on the assumption that the keiretsu affiliations are stable through time (Wu and Xu, 2005).[13]

In line with previous studies, only firms in the six largest groups[14] are considered as keiretsu members. The *Industrial Groupings in Japan* (IGJ) publication ranks company involvement in a keiretsu group on a four-point scale (weak, inclined, strong and nucleus). We present our findings, in respect to keiretsu affiliated firms, using a two-point scale and we differentiate the main banks as well.

a. *Weak:* The ratio of a group's shareholdings to total shares held by the top ten shareholders is less than 50 percent but more than 10 percent.

[12] An alternative keiretsu grouping (e.g. Sony and Toyota) is oriented about a manufacturing firm and relates to firms in the same industry with close ties as above.

[13] It should be noted that Miwa and Ramseyer (2001) emphasize classification difficulties in respect to identifying keiretsu group membership, which stem from the possibility that the keiretsu groups represent at best weak ties between firms. Notwithstanding, we follow Wu and Xu (2005) and Brown and Fung (2009) to adopt the Industrial Grouping in Japan categorization of Japanese firms into affiliated firms (and separation of these firms between distinct groups) and non-affiliated firms. We assume that the membership of each keiretsu group remains static 1990 to 2012. There were four megabank groups formed in April 2001. The number of main banks about which the keiretsu groups are said to be associated, declines from six banks to three banks in the three year period 2001-03. Nevertheless, the identity and integrity of each of the original keiretsu groups has continued even though they now also come under the new enlarged financial groupings. Our data on Japanese firm share repurchase activities suggests markedly lower levels of activity on the part of keiretsu firms relative to non-affiliated firms, which is consistent with the relative importance of ownership structure for these firms.

[14] These groups are the Mitsubishi, Mitsui, Sumitomo, Sanwa, Fuyo and DKB groups.

b. *Strong:* The ratio of a group's shareholdings to total shares held by the top ten shareholders is greater than 50 percent or the firm is a core member of the keiretsu group or a member of the group's various councils and clubs.

3.3 Cash dividend paid / received

Our principal pay out variable is cash dividend paid. Another key variable is the dividend received by keiretsu and non–keiretsu firms from a keiretsu firm.[15] In order to estimate dividends received, we must have ownership information on the dividend paying firms. The Industrial Groupings in Japan text reports 1,345 keiretsu firms across 8 keiretsu groups. About 60% (800) of these firms are publicly listed. We have identified 735 of these keiretsu listed firms in our sample across the 8 keiretsu groups. Next, in line with previous studies, we limit our analysis to the 6 main keiretsu groups and identify 633 firms in these groups in our sample. Finally, after applying the constraint of data availability in relation to the dependent and independent variables defined / discussed in table 1a, we have 555 keiretsu firms which are included in our regression analyses.

Next we identify 306 unique firms (keiretsu and non-keiretsu) which are top ten shareholders in these 555 parent keiretsu firms.[16] Therefore, we have dividend received data for 306 firms. But we use only 221 firms in our study due to the data availability of our dependent and independent variables. Of these 221 dividends receiving firms, 138 are keiretsu firms and 83 are non-keiretsu firms. In particular, 118 are net keiretsu dividend payers during at least one year from 1990 to 2012 and 73 are net keiretsu dividend receivers in at least one year from 1990 to 2012.[17]

3.4 Investments – firm level

[15] In respect to share repurchases, another possible mechanism for intra group financing, this mechanism can weaken the keiretsu linkage and hence may not – within a business group – be a preferred course of action. Share repurchases were constrained (<10% of o/s stock) prior to the enactment of the Foreign Exchange Law and the Financial System Reform Law ("Big Bang") in 1998. Keiretsu share repurchases, in our sample, are markedly lower than non-keiretsu share repurchases each year since 1996.
[16] Some firms have top-ten shareholding in more than 20 firms.
[17] 138 – 73 = 65 are Keiretsu Net Dividend payer in all sample year from 1990 to 2012; 138 – 118 = 20 are Keiretsu Net Dividend Receiver in all sample year from 1990 to 2012.

We follow Fama and French (2001) and Gopalan, Nanda and Seru (2014) to identify firm-level investment opportunities. We calculate lead aggregate 2-year investments using three different proxies – the relative annual change in book value of total assets of a firm (Fama and French, 2001, Gopalan, Nanda and Seru, 2014), the relative annual change in research and development expenditure (Pinkowitz, Stulz and Williamson, 2006) and the relative annual change in capital expenditure.

3.5 Financially distress firms

We follow Hoshi, Kashyap and Scharfstein (1990) to identify financially distressed firms in our sample. A firm in a particular year is classified as financially distressed if its operating income is greater than the firm's total interest expenses one year but less in the two subsequent consecutive years, then the second consecutive year is identified as the financially distressed year for the firm. Using the top ten shareholder data of keiretsu firms we identify a subset of financially distressed firms that are the key shareholders in a keiretsu firm and further classify them into three different pools – financially distressed firms that are a member of a same keiretsu group as the parent company in which they are a major (top ten) shareholder, financially distressed firms that are member of different keiretsu group than the parent company in which they are a major (top ten) shareholder and financially distressed non-keiretsu firms that are a major (top ten) shareholder in a keiretsu firm.

3.6 Control variables

Dividend pay out regressions

We arrange our principal control variables into groups according to their principal theoretical linkages with respect to the agency cost–based theory, the life–cycle theory and the signalling theory of corporate payout policy as well as an additional grouping of proxy variables which are included by way of an additional robustness check.

We account for the agency cost–based theory by adopting five proxy explanatory variables. First, following Dittmar and Mahrt-Smith (2007) and Pinkowitz, Stulz and Williamson (2006) we adopt cash and short term investments (CASH) as a measurement of prospective agency costs. In a similar vein, following Chay and Suh (2009) and Faccio, Lang and Young (2001) we incorporate ownership concentration (OWN) of the firm. Finally, in regard to agency costs, following Black (1976) and Jensen (1986), we adopt a leverage ratio i.e. the total debt divided by the total assets. Turning to the development of the firm's set of investment opportunities, we include in our specifications the change in total assets (Investments) following Fama and French (2001) and Denis and Osobov (2008). Also following in the line of these latter studies, we adopt the market to book ratio (MTB), to reflect the set of the firm's investment opportunities.

To account for effects related to the life-cycle theory of corporate payout policy, we adopt two proxy explanatory variables. Following DeAngelo, DeAngelo and Stulz (2006), we include a proxy explanatory variable for the phase of the life cycle of the firm, the ratio of retained earnings to total equity (RETE). In the vein of, Fama and French (2001) as well as Grullon and Michaelly (2002), we adopt the annual (percentile) market value of the firm to reflect firm size (SIZE), a complementary indication of the phase of the financial life cycle of the firm.

To control for the signalling theory of corporate payout policy we turn to a single proxy explanatory variable. We follow Wood (2001) and von Eije and Megginson (2008) and specify an earnings reporting frequency (ERF) variable, corresponding to the frequency at which earnings are reported, by a firm, per annum. A higher reporting frequency may result in a larger pay out due to greater transparency.

Furthermore, we adopt several control variables which are also expected to influence corporate pay out. Following Lintner (1956) and Miller and Rock (1985), we specify an explanatory variable corresponding to the Earnings ratio. It is computed as the earnings before interest but after tax divided by the total assets. Another key control variable, which we include in our specifications, is income risk (Chay and Suh, 2009).

Investment and financial leverage regressions

Turning to our investment and financial leverage regressions, we extend the set of control variables detailed above to include sources of financing beyond cash dividends. We account for equity financing. Specifically, we account for other proceeds from sale/issuance of stock which is the amount a company received from the sale of common and/or preferred stock. It includes amounts received from the conversion of debentures or preferred stock into common stock, sale of treasury shares and shares issued for acquisitions. We also account for long-term debt issuance. Long-term debt issuance (CF STMT) is the amount received by the company from the issuance of convertible and non-convertible long term debt, increase in capitalized lease obligations, and debt acquired from acquisitions. This is a component of our leverage ratio as defined in the appendix. Finally, we account for trade credits and total receivables in our investment regressions. Trade credits is the (net) accounts payable which represents the claims of trade creditors for unpaid goods or services which are due within the normal operating cycle of the company. Total receivables represent the amounts due to the company resulting from the sale of goods and services on credit to customers.

Summary Statistics

We report, in table 1, the summary statistics of the firm characteristics of our sample of Japanese firms, keiretsu affiliated and non-affiliated firms, keiretsu main banks and pseudo subsamples of keiretsu paying and keiretsu receiving firms, during a 26 year period, 1986 to 2012.[18] In total, we have 30,656 firm year observations consisting of 8,880 keiretsu firm years and 21,901 non-affiliated firm years. In columns 1-4 of panel A, we present the summary statistics of the firm characteristics of our full sample of Japanese firms (2,465 firms) and of subsamples of keiretsu affiliated (555 firms) and non-affiliated firms (1910 firms), and keiretsu firm main banks (6 firms - 1 bank in each keiretsu group nucleus). In columns 1-3 of panel B, we present the summary statistics of the keiretsu dividend payers (118 keiretsu firms which are net dividend payers in at least one year) and keiretsu dividend receivers (73 keiretsu firms which are net dividend receivers in at least one year), and financially distressed firms (793 firms which have experienced financial distress in at least one year).

In column 1 of panel A, we report the summary statistics of the firm characteristics of our sample of Japanese firms. Perhaps the most striking finding of table 1 panel A is reported in columns 2 and 3: the group member firms (keiretsu affiliated firms) pay out more, are relatively large in terms of market value and total assets and have higher leverage than non-affiliated firms.[19] Consistent with a predicted keiretsu cash dividend pay-out premium, keiretsu affiliated firms pay out mean cash dividends of 20.57 (median is 4.05) million US dollars annually while non-

[18] All firm characteristics are scaled by total assets except dividends (DIV), market value (MV), market-to-book (MTB), income risk, earnings reporting frequencies (ERF) and ownership concentration (OWN).

[19] This is consistent with the findings in Gopalan, Nanda and Seru (2007, 2014) in respect to typical business group firm characteristics.

affiliated firms pay out 14.21 (median is 2.44) million US dollars annually. As distinct to other group member firms in other markets keiretsu firms have low profitability (lower earnings and retained earnings to total equity than non-group members). As we consider an extensive set of firm characteristics, we can also report that our group member firms exhibit lower levels of income risk and a lower level of general investments but, interestingly, higher research and development investments than non-keiretsu firms. Unsurprisingly, keiretsu affiliated firms have higher ownership concentration than non-affiliated firms. The affiliated and non-affiliated firms are otherwise similar. They have similar amounts of cash to total assets, sales to total assets, market-to-book values and earning reporting frequencies.

We present in column 4 of panel A, the summary statistics of the six main keiretsu banks. We show that the six main banks are high dividend payers with a mean cash dividend pay out of 69.05 (median is 15.28) million US dollars annually. The main banks are large, both in terms of total assets and market capitalization and they exhibit an exceptionally high ownership concentration (the top ten shareholders own on average more than 60% of outstanding shares). Interestingly, the main banks exhibit low income risk, low levels of profitability (possibly due to low interest rates in Japan since early 1990s) and low market-to-book values. They are, nevertheless, cash rich, with high levels of retained earnings to total equity and high investment (relative to other keiretsu firms) but virtually no research and development investment.

In columns 1 and 2 of panel B, we report summary statistics of the firm characteristics of net dividend payer and receiver pseudo sub-samples of keiretsu firms, respectively. We report that keiretsu payers are relatively large in respect to total assets and market capitalisation. Further, the keiretsu dividend payers have higher earnings, higher sales and (marginally higher) levels of retained earnings to total equity. They are also relatively cash rich, with lower levels of income risk and leverage than the dividend receiver keiretsu firms. Net payers are similar to net receivers on other firm characteristics: they exhibit comparable ownership concentration, research and development investment and earnings reporting frequencies. These findings are unsurprising as they are consistent with predictions in the dividend determination literature (Allen and Michaely, 2003 and Chay and Suh, 2009). In addition, however, our finding of typical keiretsu net dividend payer and receiver firm characteristics is consistent with the possibility that cash dividends in keiretsu business groups may constitute, by design, a source of finance from relatively profitable and low income risk keiretsu firms to other keiretsu firms to assist the keiretsu firm capital budget and financial leverage policies of financially weaker firms in the group. Finally, in column 3 of panel B we report summary statistics of the firm characteristics of firms in a state of financial distress. The firms in financial distress have high accounts payable, income risk, and leverage and low earnings ratio, retained earnings to total equity and market to book values.

In panel C of table 1, we report, for net dividend receiving keiretsu affiliated firms, the pairwise correlations between our key variables of interest.[20] The set of key variables comprises dividends received, leading investment, research and development, capital expenditure and debt levels.[21] We show that keiretsu firm dividend received is correlated positively with investment (changes in total assets and research and development investments) and negatively with total debt. These results suggest that dividends received may act as a source of capital to finance these investments. The findings are pairwise correlations and hence we turn to our formal tests to assess our hypotheses.

[Please insert table 1 about here]

4. Empirical Results

In panel A of table 2, we report our findings in respect to hypothesis 1 and sub-hypotheses. We test whether firms affiliated to keiretsu groups pay more common cash dividends, than unaffiliated firms (the control group) even when accounting for the effects of well-known dividend determination variables understood to be experienced by firms. In particular, we specify a random effects panel regression model and incorporate a dummy variable, *KRF*, which takes a value of one if a firm is keiretsu affiliated and zero otherwise.

The findings reported in panel A suggest that accounting for a set of well-known dividend determination variables, keiretsu affiliated firms pay out about 22% (p-value is 0.00) more than non-affiliated firms. We use, in column 2 of panel A, a set of strength of keiretsu affiliation categories (weak and strong) to a keiretsu group to replace the keiretsu affiliation dummy

[20] Our cash dividends received data are estimated via cross shareholdings, the percentage holdings of top ten shareholders is sourced in the Industrial Groupings in Japan (2001) publication.
[21] We exclude the interest coverage ratio from the correlations analyses as we expect only the lower percentile observations of interest coverage to be negatively correlated with dividend received.

variable (KRF).[22] We include a separate dummy variable for keiretsu group main banks which are nucleus group member firms. These latter dummies will take the value one if the corresponding firm exhibits weak or strong (excluding main banks) keiretsu affiliation linkages or is a main bank of a keiretsu group, otherwise they take the value zero.

Our findings suggest that weak, strong, and main bank keiretsu affiliated firms pay out significantly more than non-affiliated firms. The influence of the keiretsu affiliation statuses on keiretsu firm pay out (relative to unaffiliated firms) rises monotonically from a reported 7.3% (p-value is 0.04) higher pay out for the weakly keiretsu affiliated firms and an 34 % (p-value is 0.02) higher pay out for the strong keiretsu affiliated firms to an 77% (p-value is 0.00) higher pay out for the nucleus keiretsu affiliated firms.[23]

In panel B of table 2, we report our findings in respect to sub-hypotheses concerning the investment opportunities of dividend receivers. It is interesting to note that there is evidence that the keiretsu pay out premium increases in the market-to-book value (C MTBV Div Rec) and lagged industry level investment (L Ind Inv Div Rec) of dividend receiving keiretsu firms. In particular, this influence is evident in the weak and strong keiretsu affiliated dividend paying firms. The additional pay out premium influences, for example due to the market-to-book value variable, are, however, relatively small at 0.013 (p-value is 0.02) and 0.02 (p-value is 0.01), respectively. It is also interesting that in respect to the keiretsu groups' main banks, the aggregate market-to-book values of dividend receiving firms do not alter the main bank keiretsu firm's dividend payout premium and nor do the receiving firm's lagged aggregate industry level investments. As a result, there is less evidence of the interest of keiretsu group main banks, as opposed to other e.g. keiretsu group nucleus members, with respect to the capital budgets of group members.

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[22] We also conduct all our tests using a more granular variation in the strength of keiretsu affiliations in the six main keiretsu groups. The results are suppressed to more concisely present our findings. Weakly affiliated firms exhibit a ratio of group shareholdings of less than 30 percent, relative to holdings of the top ten shareholders. The inclined and strong categories of keiretsu affiliation are defined similarly but with 30-50% and more than 50% of group shareholdings of the top ten share holdings, respectively. Nucleus firms are core firms, a main bank or a member of the group's various councils and clubs. Our main findings are available from the authors on request.

[23] Keiretsu group main banks show higher pay out than non-keiretsu firms in line with other keiretsu nucleus member firms. These affiliation strength related findings are in line with Faccio, Lang and Young (2001), who report that group firm dividend policies are related to the position of the firm in the group pyramid.

Turning to panel C of table 2, however, we report further findings in respect to the investment opportunities of dividend recipient firms disaggregated across strength of affiliation of the dividend recipient firm. We allow the dividend receiver status to be a member of the same keiretsu group, a member of a different keiretsu group or a non-affiliated firm. We find that the keiretsu status of the dividend recipient firm (irrespective of whether it is the same keiretsu group as the dividend paying firm), and the investment opportunities of the dividend recipient firm are positively associated with the dividend payout premium. For example, associated with the market-to-book value variable, strongly affiliated firms pay out increases by 0.036 in the market-to-book value variable of same group dividend receiving firms and by 0.015 in the market-to-book value variable of different keiretsu group dividend receiving firms. Finally, it is especially interesting to note that the investment opportunities (proxied by market-to-book values) of non-affiliated dividend receiving firms do not show a significantly positive association with the pay out of keiretsu firms, irrespective of the strength of the keiretsu affiliation.

[Please insert table 2 about here]

In table 3, we report our findings with respect to sub-hypotheses concerning an association between the payout amount decision of keiretsu affiliated firms and whether their top shareholders are in financial distress. We examine the fraction of shareholdings held by the top shareholders that are in financial distress (see panel B).[24] We find that the amount of pay out of keiretsu firms increases by 4.4% (p-value is 0.00) for each 1% rise in the top shareholdings of same group keiretsu firms, which are in financial distress. We show that the amount of pay out of keiretsu firms increases by 0.3% (p-value = 0.058) for each 1% rise in the top shareholdings of different group keiretsu firms which are in financial distress. The amount of keiretsu firm pay out is not influenced by the financial distress of top shareholders which are not a member of a keiretsu group.

A consistent finding is evident if we replace the keiretsu affiliation dummy variable (KRF), of the dividend paying firm, with weak and strong keiretsu affiliation dummies (column 2 of panel B). We show that the greater the strength of the keiretsu affiliation of the paying firm, the greater the sensitivity of the pay out amount to the financial distress of the keiretsu firm top

[24] We also repeat the analyses using a dummy variable, 1 if a top shareholder is in financial distress, 0 otherwise (see panel A).

shareholders. Specifically, in respect to weak and strong affiliated keiretsu firms for each 1% increase in the shareholding of top shareholders in financial distress there is a 1.6% and 7.8% (all p-values less than 1%) increase in the pay out amount. If keiretsu firm top shareholders of a different group are in financial distress, only strong affiliated keiretsu firms also pay out more but to a lesser extent, 0.3% (p-value = 0.00) more pay out. One striking result is that neither weak or strong affiliated keiretsu firms pay out more if top shareholders experiencing financial distress are not keiretsu group members.

An interesting exception to this monotonically increasing sensitivity of pay out amounts, across keiretsu affiliation statuses, to the financial distress of same keiretsu group top shareholders in financial distress is evident in the instance of keiretsu group main banks. Specifically, while the dividend pay out premium of the keiretsu group main bank is 1.1% (p-value = 0.00) larger when a dividend receiving firm in the same keiretsu group is in financial distress, the main bank (unlike other keiretsu firms) also exhibits an increase in its pay out premium by 1.6% (p-value = 0.00) when a non keiretsu firm, which is a dividend recipient of the main bank, is in financial distress. It is noteworthy that this latter finding is consistent with Hoshi, Kashyap and Sharfstein (1990) with respect to the lending decisions of keiretsu firm main banks with which they have strong ties, irrespective of the keiretsu affiliation of the recipient firm.

[Please insert table 3 about here]

In table 4, we present findings in respect to hypothesis 4 and sub-hypotheses. We examine the determination of leading firm level investment (the summation of the subsequent two periods' investments) – and firm level debt in table 4b. Following Hoshi, Kashyap and Scharfstein (1990) and Biddle, Hilary and Verdi (2009) and Gopalan, Nanda and Seru (2014), we include a set of suitable control variables in respect to the determination of investment and debt, in our model specification. In particular, we include in the model specification actual cash dividends received, Div Rec, by keiretsu and non-keiretsu affiliated firms, from keiretsu affiliated firms. Our model specification [3] controls for firm fixed effects and year effects, α_j and α_t, respectively.

$$\sum_{l=1}^{2} I_{i,t+l} = \alpha_0 + \partial\, Div\, Rec_{i,t} + \sum_{j=1}^{n} \beta_j\, CV_{i,t} + \alpha_j + \alpha_t + \varepsilon_t \quad [3]$$

We expect a positive (negative) association, ∂, between dividends received and lead firm level investment (debt) variables. In addition, we expect the dividend received effects on lead investment (debt) to increase with the strength of the keiretsu affiliation linkages of the dividend receiving firm. Our findings show that for a 1% increase in dividends received, lead investments increase by 0.63% (p-value: 0.04) and research and development expenditure increases by 0.4% (p-value: 0.09).[25] Our findings also show that for a 1% increase in dividends received, lead debt declines by 0.41% (p-value: 0.00) and the effect strengthens by an additional 0.28% (p-value: 0.02) when the dividend receiving firm is in financial distress. Furthermore, our findings suggest that the association between dividends received and leading firm level investments is pronounced for keiretsu firms relative to non-keiretsu firms. This association, ∂, is monotonically increasing, across strength of keiretsu affiliation of the dividend receiving firm. These associations are 0.32% (p-value 0.07) for strongly keiretsu affiliated firms and 1.04% (p-value 0.00) for nucleus affiliated firms. The research and development expenditure of nucleus keiretsu firms exhibits a 0.41% (p-value: 0.05) response to a unit percentage change in dividends received from keiretsu firms. It is interesting to note that, in contrast to typical nucleus affiliated keiretsu firms, dividends received by the keiretsu group main bank (a nucleus affiliated firm) have no significant association with the main bank's leading change in total assets, research and development expenditure and capital expenditure.

Following Almeida, Hsu and Li (2013), we test if as information asymmetries increase do dividends received become more important determinants of lead changes in total assets, research and development expenditure and capital expenditure. As information asymmetries increase, extra-group capital raising opportunities become less attractive with respect to financing future investments. We associate smaller firms with greater levels of information asymmetry (Chari, Jagannathan and Ofer, 1988). There is a negatively signed influence, of dividend received by keiretsu firms with an increase in the size of the firm, on future investment. This negative influence of the interaction variable firm size and dividend received, *Size*Div Rec*, is evident across keiretsu affiliations of dividend receiving firms in respect to the determination of both changes in total assets and research and development investment. The effect is statistically significant for strong and nucleus affiliated firms (at the 10% and 1% levels, respectively) in

[25] As a result, we identify a new determinant of investment, in the context of business group firms, relative to those determinants identified in Hoshi, Kashyap and Scharfstein (1990) and Biddle, Hilary and Verdi (2009). The capital expenditure shows no significant effects of dividend received though the direction of effect is generally as expected.

respect to changes in total assets and for nucleus firms (at the 5% level) in respect to research and development investment. This finding suggests that as relative firm size grows and information asymmetries decline the importance of dividends received in financing keiretsu firm capital budgets also declines. Therefore, the influence of dividends received on the capital budgets of keiretsu firms is not only determined by the keiretsu affiliation of the dividend receiving firm but may also be influenced by the extent of prevalent information asymmetries faced by the keiretsu dividend receiving firm with respect to the external capital markets.

[Please insert table 4 and 4b about here]

Finally, we turn to test hypothesis 5 and sub-hypotheses. We use difference-in-differences tests to investigate the influence of two types of exogenous shocks on the cash dividend pay-out of keiretsu groups in Japan: changes in the cost of external financing and changes in the taxation rate on cash dividends.

In table 5a, we present findings in respect to the dividend pay out premium, during periods of costly external financing, 1990-96. We specify interaction dummy variables with regard to this period and whether a firm is affiliated to a keiretsu group. We expect that the difference in the amount of pay out, between affiliated keiretsu and non-affiliated firms, is greater when alternative sources of financing are most expensive.[26] We control for firm fixed effects and year effects in our model specifications. We find that in the high cost of debt years, 1990-96, a keiretsu firm tends to pay out 8.8% (p-value is 0.01) more than non keiretsu affiliated firms than in low cost of debt years. Next, as postulated in hypothesis 5, we expect that the difference in cash dividend payout amounts, across keiretsu affiliated and unaffiliated firms, during the 1999 to 2003 rise in the taxation rate on cash dividends and subsequent to the 2003 reduction in the taxation rate on cash dividends is expected to decline in the former scenario and have increased in the latter. Indeed, the difference in cash dividend pay out amount is found to decline by 4.9% (p-value: .09) during the period of increased taxation rate, 1999 to 2003 and subsequent to 2003, the difference becomes significantly larger by 13.7% (p-value is 0.00). As a robustness test, we show that the relative growth of 8.8% in pay out amounts post-2003 is also evident (p-value 0.02) in the 2000-12 period. Hence, we conclude that in periods of high external borrowing costs

[26] It is perhaps noteworthy that the period of high interest rates occurs in the early 1990s which is also a period prior to the enactment of legislation which aims, in part, to dismantle keiretsu groups.

or reduced cash dividend tax rates, affiliated firms pay out more cash dividends compared to their non-affiliated counterparts.

We present findings in column 1 of table 5b in respect to difference-in-differences tests which investigate if dividend received of affiliated keiretsu influences lead investment (debt, in table 5c) during periods of high external borrowing costs more than it influences the lead investment of non-affiliated firms (hypothesis 6). In our model specification we account for a wide set of investment determinants, firm fixed effects, year effects and industry effects. It is expected that during the periods of relatively expensive borrowing costs there would be a greater impact of keiretsu firm dividends received from keiretsu firms on lead investments. We find that in high cost of debt years, a 1% increase in keiretsu firms' dividends received from keiretsu firms does impact lead investment (change in total assets) by 1.24% (p-value is 0.02) more for keiretsu affiliated firms, than for non-affiliated firms. Similarly, in column 2, we find that in high cost of debt years, 1990-96, the association of a 1% increase of dividends received on lead research and development expenditure is 0.16% (p-value is 0.00) more pronounced relative to the same influence for non-keiretsu affiliated firms. Turning to column 3, we find that in high cost of debt years, 1990-96, the association of a 1% increase of dividends received on capital expenditure is 0.94% (p-value is 0.00) more pronounced relative to the same influence for non-keiretsu affiliated firms. A similar pattern of results is evident with respect to debt in table 5c. In high cost of debt years (1990-96), dividends received have had a pronounced impact on keiretsu firm deleveraging in the order of 0.79% for a 1 % increase in dividend received, and there is an additional effect of 0.39% if the keiretsu dividend receiving firm is in financial distress. Finally, this effect becomes stronger still during the 2004-12 period when there was a marked decline in the taxation rate on cash dividends, making it a relatively attractive financing mechanism.

We also undertake difference-in-differences tests of the impact of keiretsu firms' dividend received on lead investment during periods of relatively high and low taxation rates on cash dividends. It is expected that during periods of relatively low (high) taxation on cash dividends, there would be a larger positive (negative) influence exhibited by keiretsu affiliated firms between dividends received from keiretsu affiliated firms and lead investments. In this setting, in table 5b, we show that, as per the expectation, in relatively low taxation on cash dividend periods, post-2003, the influence of a 1% increase in dividends received, across affiliated keiretsu and non-affiliated firms, on lead change in total assets is 0.87% (p-value 0.04) more pronounced for keiretsu affiliated firms. In the relatively high taxation periods, 1999-2003, we show that the association is not significantly different for keiretsu affiliated firms relative to non-affiliated firms. As a robustness test, we show that there is a relatively large 1.71% (p-value 0.00) influence, of a 1% increase in dividends received, on leading investment for keiretsu affiliated firms, relative to non-affiliated firms, post 2003 when we focus exclusively on the 2000-12 period. Turning to columns 2 and 3, we show that in the period of a high taxation rate on dividends (a 43.6% effective taxation rate), 1999 to 2003, there is no significant relative difference between the influence of dividends received on research and development and capital expenditures across keiretsu affiliated and non-keiretsu firms. In contrast, as per the expectation, there is a markedly pronounced differential, across keiretsu affiliated and non-keiretsu firms' lead research and development (0.216%; p-value is 0.00) and capital expenditure (0.939%; p-value is 0.00) responsiveness to a 1% increase in dividends received from keiretsu firms, in the 2004 to 2012 period of relatively low taxation rates on cash dividends (a 10% effective taxation rate). Hence, we conclude that in periods of high external borrowing costs or a lower cash dividend tax rate, keiretsu affiliated firms, relative to non-affiliated firms, show markedly pronounced influences of dividends received (from affiliated firms) on their capital budgets.

[Please insert table 5 a-c about here]

Finally, we report findings in table 6 in respect to hypothesis 7. For further verification, we conduct a difference-in-differences test about a period of no known potential exogenous shock, *i.e.,* we conduct falsification tests to study changes in payout amounts in keiretsu affiliated firms relative to non-affiliated firms and the influence of dividend received on firm investments. We focus on the period 2004 to 2007 inclusive – a period of a fixed dividend tax rate of 10% and prior to the financial crisis in 2008. We specify a fictional tax rate alteration year of 2005 though, in fact, there is no change in the cash dividend tax rate during this period. We find that there is no significant change in the amount of pay out in keiretsu firms, relative to non keiretsu firms, during this period. In addition, we also find that the investments and debt of keiretsu firms, relative to non-affiliated firms, do not respond differently to dividends received from keiretsu firms according to the same period.[27] As a result, these falsification tests lend support to the inference of causality, which arises from the difference-in-differences tests reported above about *de facto* potential exogenous shocks (related to the cash dividend taxation rates and the cost of extra-group financing), between the dividends received by keiretsu firms (from keiretsu firms) and the capital budgets of these keiretsu dividend recipient firms.

[**Please insert table 6 a-b about here]**

5. Conclusions

External financing is critical for firms' operation and growth, and financing through markets and financial institutions is both costly and has frictions. One alternative is financing—for instance, trade credits, through business networks, which has been shown to be an important source in emerging economies with underdeveloped markets and institutions. In this paper, we study Japan, one of the most developed countries in which some firms belong to with business groups, while others do not. We focus on cash dividends and argue that it provides a meaningful channel for keiretsu firms' investment and help these firms to reduce the debt burden.

We examine two questions using a large panel data set from the period of 1990-2012 and non-keiretsu firms as the control group. First, we are interested in whether keiretsu firms' payout policies are related to *other* keiretsu firms' financing needs. We find keiretsu firms pay more dividends than firms not affiliated with any group. The difference between the two groups of firms is greater when the dividend-receiving firms are in distress and when the linkage between them and the dividend paying firms is stronger.

Second, we study whether dividend-receiving firms' investment and leverage are related to dividends from the dividend-paying firms within the same group. We find that for keiretsu firms, a greater amount of cash dividends received is associated with higher investment and lower leverage over the next two subsequent years. This relation strengthens as the receivers' tie to the payer strengthens. Moreover, we use exogenous changes to taxes on corporate dividends and difference-in-difference tests and falsification tests (around a period when there were no tax changes), we further establish that keiretsu firms' dividend payouts have a causal impact on

[27] The same falsification test conducted over a longer period 2003-12 gives comparable results, which are available from the authors on request.

receiving firms' investment and debt policies. These results extend the literature on the role of business groups by illustrating the importance of inter-firm financing, especially during periods of high external financing costs, even for firms in a developed economy.

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Table I
Sample Overview and Summary Statistics

The table shows the summary statistics of the firm-level characteristics of our Japanese firms, keiretsu affiliated and non-affiliated firms and keiretsu main banks in Panel A. The firm-level characteristics of pseudo subsamples of keiretsu paying and keiretsu receiving firms and financially distressed firms are reported in Panel B. The pseudo subsamples of keiretsu paying and receiving firms, reported in panel B, refer to keiretsu firms which are estimated to be net dividend payers or receivers in specific firm years. Firms / firm-year observations is the total number of firms / firm–years included in the corresponding sub-sample of data. Panel C of the table reports the correlation matrix of investment, leverage, loans, bonds and dividends received data for the keiretsu firms. INV_2Yr, Debt_2Yr, Loan_2Yr and Bond_2Yr are aggregate two year lead investment and leverage variables i.e. they correspond to the summation of the subsequent two years of observations on the associated investment and debt for dividend receiving firms. The sample period is 1990 to 2012. Definitions of the firm-level characteristics are reported in appendix A1.. ***, **, and * in Panel C indicate significance at 1, 5 and 10 percent level.

Panel A: Overall Sample Summary Statistics

	(1): Japan - All Firms			(2): Keirestu Firms			(3): Non-Keirestu Firms			(4): Keirestu Banks		
	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.
DIV (M US$, 1990 prices)	19.61	2.10	139.94	24.09	3.85	114.45	18.19	1.81	147.07	112.84	103.08	780.67
TA (M US$, 1990 prices)	2185.51	383.52	9370.77	3284.49	772.16	8485.24	1837.48	319.78	9608.19	17970.64	9509.60	4387.64
MV (M US$, 1990 prices)	1129.63	166.57	5421.50	1341.34	321.23	3002.84	1062.58	138.39	5985.56	6409.20	4838.90	831.85
ER (%)	8.12	2.41	215.77	2.38	2.29	8.69	9.94	2.45	247.51	2.14	2.37	3.15
RETE (%)	31.76	50.54	607.16	37.77	45.50	673.01	29.86	52.24	584.76	48.06	51.71	27.70
MTB (%)	1.76	1.10	9.91	1.84	1.23	14.22	1.74	1.06	8.07	1.69	1.32	1.44
CASH (%)	31.00	26.99	18.89	25.99	22.54	16.07	32.59	28.68	19.44	55.95	61.42	30.28
Inc Risk	2.68	1.37	5.78	2.05	1.19	3.00	2.88	1.43	6.40	1.55	0.83	1.99
Debt_TA (%)	31.65	22.59	106.15	36.12	28.02	56.63	31.82	20.63	117.56	12.04	12.74	22.88
Loan_TA (%)	29.26	20.80	203.68	34.39	26.76	100.24	30.17	19.12	226.79	9.46	7.85	20.59
Bonds_TA (%)	2.52	0.00	27.91	2.57	0.00	26.50	2.51	0.00	28.34	2.58	0.00	6.38
ERF	2.56	2.00	1.30	2.34	2.00	1.30	2.63	2.00	1.30	2.37	2.00	1.34
OWN (%)	46.51	45.28	15.37	51.99	48.76	15.25	44.94	42.99	15.13	64.97	67.77	10.85
INV (%)	4.00	4.83	18.00	3.12	4.18	15.05	4.28	5.03	18.83	6.50	5.80	19.85
RnD_TA (%)	2.84	0.12	41.42	1.30	0.37	4.42	3.32	0.05	47.45	0.02	0.00	0.08
Sales_TA (%)	43.61	31.71	31.45	48.37	51.62	32.63	42.10	28.02	30.91	29.43	29.44	12.28
St Iss_TA (%)	0.64	0.00	21.04	0.20	0.00	2.13	0.79	0.00	24.12	0.05	0.00	0.42
T Cred_TA (%)	13.04	6.77	151.82	9.42	7.38	24.56	14.19	6.50	173.64	2.44	0.43	5.92
Firm-year observation	45652			10705			34672			275		
Firms	2847			552			2279			16		

Panel B: Keiretsu Pseudo sub-samples and Financial Distressed Firms Summary Statistics

	(1): Net Keirestu Payers			(2): Net Keirestu Receivers			(3): Financially Distressed Firms		
	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.
DIV (M US$, 1990 prices)	55.17	15.81	185.32	4.02	0.36	16.88	2.21	0.41	65.23
TA (M US$, 1990 prices)	7541.75	2664.62	13861.05	2674.22	577.66	5868.63	1355.89	231.83	7913.56
MV (M US$, 1990 prices)	2802.23	1139.88	4535.00	839.73	207.87	1892.26	814.58	104.79	4117.44
ER (%)	3.19	2.39	13.97	1.68	1.88	7.48	-3.46	1.19	116.24
RETE (%)	42.24	47.72	106.15	66.22	31.45	1319.06	-11.66	8.82	433.31
MTB (%)	1.71	1.40	1.93	2.08	1.31	27.47	1.88	0.99	10.16
CASH (%)	23.81	20.54	15.05	26.31	21.68	17.50	29.09	24.68	23.16
Inc Risk	1.64	0.95	2.85	2.81	1.85	3.28	14.10	12.26	7.28
Debt_TA (%)	37.47	30.77	98.99	37.49	36.36	29.69	49.70	48.84	100.56
Loan_TA (%)	34.71	27.91	98.96	35.34	34.08	29.28	44.86	42.11	406.20
Bonds_TA (%)	2.77	0.13	5.55	2.15	0.00	5.27	5.35	0.00	26.15
ERF	2.35	2.00	1.32	2.20	2.00	1.28	2.82	4.00	1.28
OWN (%)	47.37	42.64	15.37	53.21	50.98	15.43	49.55	49.78	16.29
INV (%)	4.03	5.42	14.35	1.24	2.52	15.82	3.84	4.31	19.52
RnD_TA (%)	1.35	0.62	1.77	1.02	0.30	1.67	1.41	0.20	18.90
Sales_TA (%)	51.86	62.52	33.01	48.97	53.53	32.13	25.10	43.79	29.44
St Iss_TA (%)	0.18	0.00	1.31	0.16	0.00	1.85	0.73	0.00	12.22
T Cred_TA (%)	7.98	7.41	9.85	6.68	5.40	10.89	28.99	6.98	355.34
Firm-year observation	3222			1132			3114		
Firms	195			124			1396		

Panel C: Correlation Matrix between Dividend Received and Future firm-level Investment and Leverage of Keiretsu Firms

	DR	INV_2Yr	LR_2Yr	Loan_2Yr	Bond_2Yr
DR	1				
INV_2Yr	0.065**	1			
Debt_2Yr	-0.034**	-0.020**	1		
Loan_2Yr	-0.034***	-0.014**	0.580***	1	
Bond_2Yr	-0.004**	0.015	0.032***	0.436***	1

Table II
Keiretsu Dividend Payout Premium and Dividend Receiver Investment Opportunity

The table presents tests of keiretsu payout premia across the strength of keiretsu affiliations and the keiretsu main banks (Panel A). It also reports tests of whether the payout premia are influenced by the current investment opportunities of dividend receiving firms (Panel B) and if this influence is related to the same group (S_KRF) or different group (D_KRF) keiretsu or non-keiretsu (N_KRF) firm status of the dividend receiving firm (panel C). The dependent variable is the natural logarithm of the total cash dividend paid, US$ real millions (1990 prices). The keiretsu affiliation status of dividend paying firms varies between weak, strong and main bank firms. The market-to-book value (MTB) of the dividend receiving firm (DR) is presented contemporaneously ‘C’. ‘C_MTB_DR’ is the summation of contemporary market-to-book value of all the dividend receiving firms of a specific dividend payer. The lagged mean industry level investment growth of the dividend receiving firms is indicated as ‘L_Ind_Inv_DR’. C_MTB_DR is also disaggregated between same keiretsu group (C_MTB_DR_S_KRF), different keiretsu group (C_MTB_DR_D_KRF) and non-keiretsu (C_MTB_DR_N_KRF) firms. We control for firm level – Size, ER, RETE, MTB, CASH, Inc Risk, Loans_TA, Debt_TA, ERF, INV, OWN, RnD_TA, Sales_TA, St Iss_TA and T Cred_TA. The time-varying independent variables are lagged by a one-year period. Definitions of the reported independent variables and control variables (suppressed) are presented in appendix A1. The sample period is 1990–2012. For these tests we run random effects panel regressions and standard errors are clustered at firm level. We control for year and industry fixed effects. ***, **, and * indicate significance at 1, 5 and 10 percent level.

	Panel A		Panel B		Panel C	
	(1)	(2)	(1)	(2)	(1)	(2)
KRF	0.196***		0.178***		0.176***	
	(0.037)		(0.038)		(0.038)	
Weak		0.074**		0.010**		0.010**
		(0.036)		(0.005)		(0.005)
Strong		0.234***		0.206***		0.220***
		(0.057)		(0.059)		(0.064)
Bank		0.621**		0.600*		0.589*
		(0.267)		(0.348)		(0.352)
C_MTB_DR_S_KRF * KRF					0.023***	
					(0.002)	
C_MTB_DR_D_KRF * KRF					0.004***	
					(0.001)	
C_MTB_DR_N_KRF * KRF					0.001	
					(0.002)	
C_MTB_DR * KRF			0.010**			
			(0.005)			
L_Ind_Inv_DR * KRF			0.004***		0.003***	
			(0.001)		(0.001)	
C_MTB_DR_S_KRF * Weak						0.015***
						(0.004)
C_MTB_DR_D_KRF * Weak						0.004
						(0.003)
C_MTB_DR_N_KRF * Weak						0.000
						(0.001)
C_MTB_DR * Weak				0.023***		
				(0.002)		
L_Ind_Inv_DR * Weak				0.003***		0.003***
				(0.001)		(0.001)
C_MTB_DR_S_KRF * Strong						0.046***
						(0.017)
C_MTB_DR_D_KRF * Strong						0.014**
						(0.564)
C_MTB_DR_N_KRF * Strong						-0.019

						(0.018)
C_MTB_DR * Strong				0.123***		
				(0.005)		
L_Ind_Inv_DR * Strong				0.004***		0.004***
				(0.001)		(0.001)
C_MTB_DR_S_KRF * Bank						0.059***
						(0.015)
C_MTB_DR_D_KRF * Bank						0.041
						(0.033)
C_MTB_DR_N_KRF * Bank						-0.037
						(0.038)
C_MTB_DR * Bank				0.001		
				(0.001)		
L_Ind_Inv_DR * Bank				-0.003		-0.003
				(0.003)		(0.003)
Constant	-0.203***	-0.198***	-0.202***	-0.197***	-0.203***	-0.197***
	(0.030)	(0.030)	(0.030)	(0.030)	(0.030)	(0.030)
Observations	42950	42950	42950	42950	42950	42950
No. of Firms	2847	2847	2847	2847	2847	2847
R sq. within	0.1991	0.1992	0.2001	0.2003	0.2002	0.2005
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes
Regression Type	Random	Random	Random	Random	Random	Random

Table III
Keiretsu Dividend Payout Premium and Dividend Receiver Financial Distress

The table presents tests of an association between the amount of pay out of keiretsu firms and the financial distress of dividend receiving firms, their strength of affiliation with the keiretsu group of the paying firm and the strength of affiliation of the keiretsu paying firm. The dependent variable is the natural logarithm of the total cash dividend paid, US$ real millions (1990 prices). The keiretsu affiliation status of dividend paying firms varies between weak, strong and main-bank firms. In panel A, our interaction variables are dummy variables. They take a value of 1 if a top shareholder of the keiretsu dividend payer is in financial distress and if the financially distressed top shareholder is a keiretsu same group member (S_KRF_Dist.) or a different keiretsu group member (D_KRF_Dist.) or an unaffiliated firm (N_KRF_Dist.) to a keiretsu group. The interaction variables also allow for the strength of affiliation of the keiretsu dividend payer. In panel B, our interaction variable contains a summation of fractions of shareholdings of financially distressed shareholder in the keiretsu paying firm instead of a dummy variable with respect to strength of affiliation. We control for firm level – Size, ER, RETE, MTB, CASH, Inc Risk, Debt_TA, Loans_TA, ERF, INV, RnD_TA, Sales_TA, St Iss_TA and T Cred_TA. All the time-varying independent variables are lagged by one-year period. The sample period is 1990 to 2012. Definitions of the reported independent variables and control variables (suppressed) are presented in appendix A1. For these tests we run fixed effects panel regressions and standard errors are clustered at firm level. We control for year fixed effects. ***, **, and * indicate significance at 1, 5 and 10 percent level.

	Panel A		Panel B	
	(1)	(2)	(1)	(2)
KRF * S_KRF_Dist.	0.028***		0.016***	
	(0.002)		(0.002)	
KRF * D_KRF_Dist.	0.001**		0.012**	
	(0.001)		(0.006)	
KRF * N_KRF_Dist.	0.012		0.001	
	(0.009)		(0.009)	
Weak * S_KRF_Dist.		0.073		0.012
		(0.044)		(0.006)
Weak * D_KRF_Dist.		0.008		0.001
		(0.078)		(0.002)
Weak * N_KRF_Dist.		-0.185		-0.039
		(0.120)		(0.028)
Strong * S_KRF_Dist.		0.028***		0.011
		(0.002)		(0.003)
Strong * D_KRF_Dist.		0.012**		0.005***
		(0.006)		(0.002)
Strong * N_KRF_Dist.		-0.060		0.003
		(0.126)		(0.004)
Bank * S_KRF_Dist.		0.013***		0.015***
		(0.005)		(0.004)
Bank * D_KRF_Dist.		N/A		N/A
Bank * N_KRF_Dist.		0.057		0.002*
		(0.072)		(0.001)
Constant	0.174***	0.174***	0.174***	0.173***
	(0.039)	(0.039)	(0.039)	(0.039)
Observations	42950	42950	42950	42950
No. of Firms	2847	2847	2847	2847
R sq. within	0.2033	0.2034	0.2193	0.2194
Year Fixed Effect	Yes	Yes	Yes	Yes
Regression Type	Fixed	Fixed	Fixed	Fixed

Table IV (a)
Firm-level Future Investments, RnD and Capital Expenses and contemporary Dividend Received

The table reports tests of associations between dividends received and same firm-level leading investment, debt, loans, and, bonds, accounting for the strength of keiretsu affiliation of the dividend receiving firm and whether the firm is in financial distress. The dependent variable is the sum of (t+1) and (t+2) firm investments in Panel A, the sum of (t+1) and (t+2) firm debts scaled by total assets in Panel B, the sum of (t+1) and (t+2) firm loans scaled by total assets in Panel C and the sum of (t+1) and (t+2) firm bonds scaled by total assets in Panel D. The keiretsu affiliation status of dividend receiving firms varies between weak, strong and main-bank firms. Dist._Yr. is the firm-level annual dummy variable, equal to one if the dividend receiving firm is in financial distress in a specific year, otherwise zero. We control for firm level – Size, ER, RETE, MTB, CASH, Inc Risk, Debt_TA, ERF, INV, Sales_TA, St Iss_TA and T Cred_TA. All the time-varying independent variables are lagged by one-year period. The sample period is 1990 to 2012. We use the natural logarithm of the firm-specific proxy variables which are preceded by 'Ln_'. Definitions of the reported independent variables and control variables (suppressed) are presented in appendix A1. Standard errors are clustered at firm level. We control for year fixed effects. ***, **, and * indicate significance at 1, 5 and 10 percent level.

	Panel A: INV_2Yr			Panel B: Debt_2Yr			Panel C: Loan_2Yr			Panel D: Bonds_2Yr		
	KRF Only	All Firms		KRF Only	All Firms		KRF Only	All Firms		KRF Only	All Firms	
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
KRF		1.337**			0.634***			0.321***			0.075**	
		(0.551)			(0.199)			(0.104)			(0.036)	
KRF * Dist._Yr.		0.409**			0.242**			0.057**			0.296	
		(0.192)			(0.096)			(0.024)			(0.482)	
Ln_DR	0.024**		0.084***	-0.099***		-0.020***	-0.029**		-0.027**	-0.110***		-0.098***
	(0.010)		(0.011)	(0.012)		(0.011)	(0.012)		(0.013)	(0.037)		(0.033)
Ln_DR * Dist._Yr.	0.057*			-0.077***			-0.310***			-0.096**		
	(0.029)			(0.011)			(0.059)			(0.038)		
Ln_DR * Weak			0.096			-0.026*			-0.259**			-0.005
			(0.327)			(0.016)			(0.121)			(0.004)
Ln_DR * Strong			0.031**			-0.514***			-0.385**			-0.126**
			(0.015)			(0.099)			(0.158)			(0.060)
Ln_DR * Bank			0.827			-0.554			-0.049			-0.505*
			(1.175)			(0.644)			(0.528)			(0.299)
Ln_DR * Dist._Yr. * KRF		0.074***			-0.035***			-0.170***			-0.052*	
		(0.019)			(0.009)			(0.029)			(0.028)	
Ln_DR * Dist._Yr. * Weak			0.228*			-0.143**			-0.113			0.676
			(0.131)			(0.062)			(0.093)			(1.021)

Ln_DR * Dist._Yr. * Strong			0.410**			-1.811**			-0.378***			-0.255**
			(0.191)			(0.765)			(0.012)			(0.107)
Ln_DR * Dist._Yr. * Bank			0.565**			-0.312			-0.214*			-0.183
			(0.291)			(0.609)			(0.119)			(0.115)
Debt_TA	-0.025***	-0.036***	-0.035***	1.199***	1.710***	1.245***	1.193***	1.690***	1.240***	0.006	0.006	0.005
	(0.002)	(0.013)	(0.003)	(0.086)	(0.046)	(0.110)	(0.084)	(0.037)	(0.109)	(0.005)	(0.007)	(0.004)
Constant	-1.432	-1.174	-3.343	39.171***	10.926***	34.774***	39.686***	13.540***	35.303***	-0.515	-1.850	-0.529
	(5.121)	(2.105)	(5.040)	(5.995)	(2.208)	(5.536)	(4.873)	(2.028)	(4.841)	(2.711)	(1.195)	(2.399)
Observations	4285	5392	5392	4285	5392	5392	4285	5392	5392	4285	5392	5392
No. of Firms	221	277	277	221	277	277	221	277	277	221	277	277
R sq. within	0.0212	0.0143	0.0204	0.615	0.6096	0.6236	0.612	0.6045	0.6218	0.0533	0.0531	0.057
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Regression Type	Fixed	Random	Fixed	Fixed	Random	Fixed	Fixed	Random	Fixed	Fixed	Random	Fixed

Table IV (b)
Firm-level Future Investments and contemporary Bank Loans

The table reports tests of associations between leading same firm-level investment and loans received, accounting for the strength of keiretsu affiliation of the dividend receiving firm and whether the firm is in financial distress. The sample period is 1990 to 2012. The dependent variable is the sum of (t+1) and (t+2) firm investment. The keiretsu affiliation status of dividend receiving firms varies between weak, strong and main-bank firms. Dist._Yr. is the firm-level annual dummy variable, equal to one if the loan receiving firm is in financial distress in a specific year, otherwise zero. We control for firm level – Size, ER, RETE, MTB, CASH, Inc Risk, Debt_TA, ERF, INV,, RnD_TA, Sales_TA, St Iss_TA and T Cred_TA. All the time-varying independent variables are lagged by one-year period. Definitions of the reported independent variables and control variables (suppressed) are presented in appendix A1. Standard errors are clustered at firm level. We control for year fixed effects. ***, **, and * indicate significance at 1, 5 and 10 percent level.

	INV_2Yr		
	KRF Only	All Firms	
	(1)	(2)	(3)
KRF		1.335***	
		(0.110)	
KRF * Dist._Yr.		0.369*	
		(0.192)	
Loan_TA	-0.193**		-0.274***
	(0.080)		(0.103)
Loan_TA * Dist._Yr.	0.001		
	(0.006)		
Loan_TA * Weak			-0.105
			(0.115)
Loan_TA * Strong			-0.101
			(0.078)
Loan_TA * Bank			-0.044
			(0.095)
Loan_TA * Dist._Yr. * KRF		0.007	
		(0.006)	
Loan_TA * Dist._Yr. * Weak			0.012
			(0.157)
Loan_TA * Dist._Yr. * Strong			-0.004
			(0.006)
Loan_TA * Dist._Yr. * Bank			0.111
			(0.072)
Debt_TA	-0.166**	-0.037***	-0.147*
	(0.080)	(0.014)	(0.080)
Constant	-1.553	-1.091	-1.735
	(5.103)	(2.108)	(4.900)
Observations	4285	5392	5392
No. of Firms	221	277	277
R sq. within	0.022	0.014	0.023
Year Fixed Effect	Yes	Yes	Yes
Regression Type	Fixed	Random	Fixed

Table V (a)
Difference-in-Differences Tests of the Keiretsu Firm Payout Premium in respect to External Financing Costs and Cash Dividend Taxation Rates

The table reports tests of the impact on the keiretsu payout premium during periods of relatively expensive external financing i.e. high cost of borrowing years (HCBY_1990-96) and periods of high (TY_1999-03) taxation rates on common cash dividends. The sample periods examined are 1990 to 2012 and 2000 to 2012. The dependent variable is the natural logarithm of the total cash dividend paid, US$ real millions (1990 prices). Definitions of 'C_MTB_DR_S_KRF' and 'L_IND_INV' are outlined in table II and 'S_KRF_Dist.' in table III. We control for firm level – Size, ER, RETE, MTB, CASH, Inc Risk, Debt_TA, Loans_TA, ERF, INV, RnD_TA, Sales_TA, St Iss_TA and T Cred_TA. All the time-varying independent variables are lagged by one-year period. Definitions of the reported independent variables and control variables (suppressed) are presented in appendix A1. For these tests we run fixed effects panel regressions and standard errors are clustered at firm level. We control for year fixed effects. ***, **, and * indicate significance at 1, 5 and 10 percent level.

	(1): High cost of borrowing years	(2): Dividend tax change year	
	1990-2012	1990-2012	2000-2012
HCBY_1990–96 * KRF	0.096**		
	(0.041)		
HCBY_1990–96 * KRF * C_MTB_DR_S_KRF	0.067***		
	(0.009)		
HCBY_1990–96 * KRF * L_Ind_Inv_DR	0.001		
	(0.027)		
HCBY_1990–96 * KRF * S_KRF_Dist.	0.005***		
	(0.002)		
TY_2004–12 * KRF		0.192***	0.407***
		(0.066)	(0.072)
TY_2004–12 * KRF * C_MTB_DR_S_KRF		0.111***	0.126***
		(0.042)	(0.039)
TY_2004–12 * KRF * L_Ind_Inv_DR		0.002	-0.005
		(0.002)	(0.005)
TY_2004–12 * KRF * S_KRF_Dist.		0.012**	0.004**
		(0.006)	(0.002)
Constant	0.142***	0.176***	0.083**
	(0.039)	(0.039)	(0.041)
Observations	42950	42950	30318
No. of Firms	2847	2847	2776
R sq. within	0.207	0.205	0.196
Year Fixed Effect	Yes	Yes	Yes
Regression Type	Fixed	Fixed	Fixed

Table V (b)
Difference-in-Differences Tests of the Keiretsu Firm-level Future Investments, RnD and Capital Expenses with respect to External Financing Costs and Cash Dividend Taxation Rates

The table reports a test of the impact of dividends received, across affiliated keiretsu and non-keiretsu firms, on lead investment during periods of high external borrowing costs (HCBY_1990-96) and periods of low (TY_2004-12) taxation rates on common cash dividends. The table also reports tests with respect to whether there is association between future investments and contemporaneous debt-level of the dividend receiving firm. The sample periods examined are 1990 to 2012 and 2000 to 2012. The dependent variable is the sum of (t+1) & (t+2) firm investments in Panel A, the sum of (t+1) & (t+2) firm debt scaled by total assets in Panel B, the sum of (t+1) & (t+2) firm loans scaled by total assets in Panel C and the sum of (t+1) & (t+2) bonds scaled by total assets in Panel D. Definition of 'Dist._Yr.' is outlined in table IV. We control for firm level – Size, ER, RETE, MTB, CASH, Inc Risk, Debt_TA, ERF, INV, RnD_TA, Sales_TA, St Iss_TA and T Cred_TA. All the time-varying independent variables are lagged by one-year period. We use the natural logarithm of the firm-specific proxy variables preceded by 'Ln_'. Definitions of the reported independent variables and control variables (suppressed) are presented in appendix A1. For these tests we run fixed effects panel regressions and standard errors are clustered at firm level. We control for year fixed effects. ***, **, and * indicate significance at 1, 5 and 10 percent level.

	Panel A: INV_2Yr			Panel B: Debt_2Yr			Panel C: Loan_2Yr			Panel D: Bonds_2Yr		
	(1): High cost of borrowing year	(2): Dividend tax change year		(1): High cost of borrowing year	(2): Dividend tax change year		(1): High cost of borrowing year	(2): Dividend tax change year		(1): High cost of borrowing year	(2): Dividend tax change year	
	1990-2012	1990-2012	2000-2012	1990-2012	1990-2012	2000-2012	1990-2012	1990-2012	2000-2012	1990-2012	1990-2012	2000-2012
Ln_DR	0.109***	0.082***	0.252***	-0.049***	-0.108***	-0.231***	-0.083***	-0.137***	-0.147***	-0.034***	-0.090***	-0.083***
	(0.012)	(0.034)	(0.029)	(0.009)	(0.039)	(0.044)	(0.007)	(0.038)	(0.040)	(0.005)	(0.005)	(0.007)
HCBY_1990–96 * Ln_DR	0.227***			-0.104**			0.084			0.020		
	(0.062)			(0.044)			(0.060)			(0.028)		
HCBY_1990–96 * Ln_DR * KRF	0.132			-0.111***			-0.103*			-0.008		
	(0.109)			(0.043)			(0.062)			(0.033)		
HCBY_1990–96 * Ln_DR * KRF * Dist._Yr.	0.460***			-1.570***			-2.542***			-0.065**		
	(0.121)			(0.611)			(0.613)			(0.031)		
HCBY_1990–96 * Ln_DR * KRF * C_MTB2	0.074**											
	(0.029)											
TY_2004–12 * Ln_DR		0.053**	0.080***		0.020	-0.026		-0.007	-0.023		-0.009	-0.003
		(0.023)	(0.022)		(0.016)	(0.030)		(0.026)	(0.028)		(0.008)	(0.006)
TY_2004–12 * Ln_DR * KRF		0.025**	0.022		-0.157***	0.008		-0.029**	-0.322***		0.015	-0.007
		(0.012)	(0.014)		(0.026)	(0.020)		(0.014)	(0.125)		(0.011)	(0.006)
TY_2004–12 * Ln_DR * KRF * Dist._Yr.		0.064***	0.045***		-1.583***	-1.153*		-2.567***	-1.652**		-0.008*	0.004
		(0.009)	(0.007)		(0.622)	(0.697)		(0.623)	(0.702)		(0.005)	(0.005)

TY_2004–12 * Ln_DR * KRF * C_MTB2		0.021**	0.024**									
		(0.008)	(0.009)									
Debt_TA	-0.035	-0.035	-0.039	1.245***	1.245***	0.578*	1.240***	1.240***	0.587*	0.005	0.006	-0.009
	(0.030)	(0.030)	(0.067)	(0.110)	(0.110)	(0.314)	(0.109)	(0.109)	(0.320)	(0.004)	(0.004)	(0.010)
Constant	-3.460	-2.415	-16.685*	34.736***	34.696***	55.883***	35.410***	35.541***	51.032***	-0.674	-0.845	4.851*
	(4.982)	(5.070)	(8.894)	(5.602)	(5.635)	(15.412)	(4.800)	(4.864)	(15.615)	(2.503)	(2.497)	(2.737)
Observations	5392	5392	3110	5392	5392	3110	5392	5392	3110	5392	5392	3110
No. of Firms	277	277	268	277	277	268	277	277	268	277	277	268
R sq. within	0.0225	0.0204	0.1395	0.6236	0.6235	0.3532	0.6218	0.6218	0.3622	0.0549	0.0551	0.0404
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Regression Type	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Table V (c)
Difference-in-Differences Tests of the Keiretsu Firm-level Future Investments and Bank Loan received with respect to External Financing Costs and Cash Dividend Taxation Rates

The table reports tests of the impact on the keiretsu payout premium during periods of relatively expensive external financing i.e. high cost of borrowing years (HCBY_1990-96) and years of low (TY_2004-12) taxation rates on common cash dividends. The sample periods examined are 1990 to 2012 and 2000 to 2012. The dependent variable is the natural logarithm of the total cash dividend paid, US$ real millions (1990 prices). Definitions of 'C_MTB_DR_S_KRF' and 'L_IND_INV' are outlined in table II and 'S_KRF_Dist.' in table III. We control for firm level – Size, ER, RETE, MTB, CASH, Inc Risk, Debt_TA, ERF, INV,, RnD_TA, Sales_TA, St Iss_TA and T Cred_TA. All the time-varying independent variables are lagged by one-year period. Definitions of the reported independent variables and control variables (suppressed) are presented in appendix A1. For these tests we run fixed effects panel regressions and standard errors are clustered at firm level. We control for year fixed effects. ***, **, and * indicate significance at 1, 5 and 10 percent level.

	(1) : High cost of borrowing years	(2): Dividend tax change year	
	1990-2012	1990-2012	1990-2012
Loan	-0.226***	-0.223***	-0.271***
	(0.077)	(0.074)	(0.066)
HCBY_1990–96 * Loan_TA	0.046		
	(0.032)		
HCBY_1990–96 * Loan_TA * KRF	-0.030		
	(0.049)		
HCBY_1990–96 * Loan_TA * KRF * Dist._Yr.	-0.043		
	(0.071)		
TY_2004–12 * Loan_TA		-0.039	-0.052
		(0.046)	(0.061)
TY_2004–12 * Loan_TA * KRF		-0.041	0.024
		(0.055)	(0.064)
TY_2004–12 * Loan_TA * KRF * Dist._Yr.		0.132***	0.061
		(0.042)	(0.040)
Debt_TA	-0.179**	-0.148*	-0.211*
	(0.077)	(0.077)	(0.100)
Constant	-3.519	-1.326	-16.633
	(5.064)	(5.205)	(9.377)
Observations	5392	5392	3110
No. of Firms	277	277	268
R sq. within	0.0218	0.0277	0.1454
Year Fixed Effect	Yes	Yes	Yes
Regression Type	Fixed	Fixed	Fixed

Table VI
A difference-in-differences falsification test about a fictional exogenous shock in 2006-07

The table reports a test of whether there is (1) a change in the amount of keiretsu pay out and (2) a change in the impact of dividend received with respect to keiretsu two-year lead aggregate investments, debt, loans and bonds in the 2006 to 2007 period relative to the 2004 to 2005 period. The year of the fictional exogenous shock (SY_2006-07) is 2006. The dependent variable is the natural logarithm of the total cash dividend paid, US$ real millions (1990 prices) in Panel A, sum of (t+1) & (t+2) firm investments in Panel B, sum of (t+1) & (t+2) firm Debt scaled by total assets in Panel C, sum of (t+1) & (t+2) firm Loans scaled by total assets in Panel D and sum of (t+1) & (t+2) firm Bonds scaled by total assets in Panel E. The sample period is 2004 to 2007. We control for firm level – Size, ER, RETE, MTB, CASH, Inc Risk, Debt_TA, ERF, INV,, RnD_TA, Sales_TA, St Iss_TA and T Cred_TA. All the time-varying independent variables are lagged by one-year period. We use the natural logarithm of the firm-specific proxy variables preceded by 'Ln_'. Definitions of the reported independent variables and control variables (suppressed) are presented in appendix A1. For these tests we run fixed effects panel regressions and standard errors are clustered at firm level. We control for year fixed effects. ***, **, and * indicate significance at 1, 5 and 10 percent level.

	Panel A: Ln_DIV	Panel B: INV_2Yr	Panel C: Debt_2Yr	Panel D: Loan_2Yr	Panel E: Bond_2Yr
SY_2006-07 * KRF	0.185	1.029***	1.258	0.237	1.021
	(0.139)	(0.126)	(1.111)	(1.325)	(0.895)
Ln_DR		-0.045	-0.054	-0.047	-0.007
		(0.042)	(0.047)	(0.047)	(0.027)
SY_2006–07 * Ln_DR		0.100***	0.034	0.032	0.002
		(0.018)	(0.022)	(0.022)	(0.011)
SY_2006–07 * Ln_DR * KRF		0.211	-0.009	-0.017	0.007
		(0.185)	(0.030)	(0.031)	(0.010)
Constant	0.493***	-16.849	93.481***	87.084***	6.397
	(0.057)	(16.482)	(8.891)	(8.686)	(4.360)
Observations	9812	975	975	975	975
No. of Firms	2649	251	251	251	251
R sq. within	0.081	0.401	0.099	0.093	0.070
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes
Regression Type	Fixed	Fixed	Fixed	Fixed	Fixed

Appendix A1
List of variables and their definitions

Variables / acronym	Definition
Cash dividend paid – DIV	Total real amount distributed as the cash dividend by the firm, in million US$, 1990 prices.
Market value – MV	Year-end market capitalization of the firm, in million US$, 1990 prices.
Size	Annual percentile ranking of the firm based on year-end market value.
Total Assets – TA	Year-end total assets of the firm, in million US$, 1990 prices.
Earnings ratio – ER	Earnings before interest but after tax divided by total assets, in percentage.
RETE	Retained earnings divided by total value of firm equity, in percentage.
MTB	Market value of equity divided by book value of equity.
CASH	Cash and short term investments divided by total assets, in percentage.
Inc Risk	Standard deviation of ratio of past five years net income to total assets
Leverage ratio – Debt_TA	Total long–term and short–term debt divided by total assets, in percentage. Debt contains loans, bonds o/s, other inter-corporate borrowing and debentures.
Loans – Loan_TA	Total debt borrowed in loan form divided by total assets, in percentage.
Other borrowing – Bonds_TA	Total inter-corporate borrowing, bonds outstanding and debentures, divided by total assets, in percentage.
ERF	Earning reporting frequency, which takes the value 1, 2, 3 and 4, representing the number of times earnings are reported annually.
Ownership concentration – OWN	Number of shares held by ten largest shareholders as a fraction of the total number of shares outstanding, in percentage.
Investment – INV	Relative annual change in total assets of a firm, in percentage.
R&D expense – RnD_TA	Research and development expenditures divided by total assets, in percentage.
Sales_TA	Total sales divided by total assets, in percentage
Stock issue – St Iss_TA	Total proceeds received from the sale of common and / or preferred shares divided by total assets, in percentage.
Total Credit – T Cred_TA	Difference of total accounts payable and receivable divided by total assets, in percentage.
Keiretsu – KRF	Dummy variable that takes the value of 1 if the firm belongs to one of the six Keiretsu groups – Mitsubishi, Mitsui, Sumitomo, Sanwa, Fuyo and DKB.
Weak	Dummy variable that takes the value of 1 if the firm belongs to one of the six Keiretsu groups and the ratio of group's shareholdings to total shares held by the top ten shareholders is less than 50 percent but more than 10 percent.
Strong	Dummy variable that takes the value of 1 if the firm belongs to one of the six Keiretsu groups and the ratio of group's shareholdings to total shares held by the top ten shareholders is more than 50 percent or the firm is the core member of the Keiretsu group or a member of the group's various councils and clubs.
Bank	Dummy variable that takes the value of 1 if the firm is one of the six Keiretsu main banks.
Distress firm-year – Dist._Yr.	Firm–year specific dummy variable that takes the value of 1 for a particular firm in year t for firm whose operating income was greater than its total interest payments in year t–2, but less in the next two years t–1 & t.
Same Group Distress – S_KRF_Dist.	Financially distressed firm which is a shareholder of the Keiretsu firm and belongs to the same Keiretsu group. We use both dummy and percentage of shareholding.
Diff. Group Distress – D_KRF_Dist.	Financially distressed firm which is a shareholder of the Keiretsu firm but belongs to the different Keiretsu group. We use both dummy and percentage of shareholding.
Non–Keiretsu Distress– N_KRF_Dist.	Financially distressed firm which is a shareholder of the Keiretsu firm but it is not a Keiretsu firm. We use both dummy and percentage of shareholding.
Dividend received – DR	Dividend received by a Keiretsu and Non–Keiretsu firms from a Keiretsu firm, in million US$, 1990 prices. We identify the name and percentage shareholding of top ten shareholders for 306 Keiretsu firms from "Industrial Groupings in Japan: The Anatomy of the Keiretsu"(Brown & Company 1999, 2001). We calculate the dividend received by each top ten shareholder by multiplying the dividend paid by the parent firm and the percentage holding of the shareholder in the parent firm. Next we aggregate the dividend received

	wherever the firm is a top ten shareholder in more than one Keiretsu firm. Finally assuming that the percentage shareholding is static overtime (Chemmanur et al., 2011 and Faccio et al., 2001) and using the mid–point of our study period horizon we replicate this strategy for all the twenty–three years (1990–2012).
Net Keiretsu Payer/ Receiver	Dividend received data is recorded as firm year specific observations. When dividends paid by a firm exceed estimates of dividends received we identify a Net Keiretsu Payer and when dividends paid by a firm are less than estimates of dividends received, we identify Net Keiretsu Receivers. It is thus likely that estimates of Net Keiretsu Payers are biased upwards and estimates of Net Keiretsu Receivers are biased downwards.
HCBY_1990-96	Dummy variable that takes the value of 1 for the period of high cost of borrowing years i.e. from 1990 to 1996, when the Japanese corporate bond yield was more than 2.5%.
TY_1999–03	Dummy variable that takes the value of 1 for the years 1999 to 2003 when the cash dividend tax rate was 43.6%.
TY_2004-12	Dummy variable that takes the value of 1 for the years 2004 to 2011 when the cash dividend tax rate was 10.0%
SY_2006–07	Dummy variable that takes the value of 1 for the years 2006 and 2007 when there was no change in the cash dividend tax or exogenous shock for the falsification test.
INV_2Yr	Sum of the subsequent two years of investment (INV) for year t+1 & t+2, stated in %.
Debt_2Yr	Sum of the subsequent two years of debt (Debt_TA) for year t+1 & t+2, stated in %.
Loan_2Yr	Sum of the subsequent two years of bank loans (Loan_TA) for year t+1 & t+2, stated in %.
Bond_2Yr	Sum of the subsequent two years of inter-corporate borrowing (Bonds_TA) for year t+1 & t+2, stated in %.
C_MTB2	Contemporary market to book value of the dividend receiving firm.
Ind_Inv_DR	Aggregate industry-level investment growth of the dividend receiving firm.
C_MTB_DR	Summation of contemporary market-to-book value of all the dividend receiving firms of a specific dividend payer.
C_MTB_DR_S_KRF	Total contemporary market-to-book value of all the dividend receiving firms in the same Keiretsu group of a specific Keiretsu dividend payer.
C_MTB_DR_D_KRF	Total contemporary market-to-book value of all the dividend receiving firms in the different Keiretsu group of a specific Keiretsu dividend payer.
C_MTB_DR_N_KRF	Total contemporary market-to-book value of all the non-keiretsu dividend receiving firms from a Keiretsu dividend payer.